

Figure 9: City pie chart



Figure 10: Adding more sub-layers

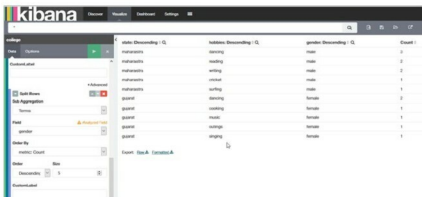


Figure 11: Data table

Visualising data

In the previous section, we used Elasticsearch to store, search and analyse the data. But the actual power comes with the visualisation of data. Assuming your Elasticsearch and Kibana are up, open a browser, enter the URL `127.0.0.1:5601` and press **Enter**. In the text box, enter 'College' as we are going to use our college index for

visualisation. Then click on the 'Create' button as shown in Figure 7.

On the top bar, click on 'Visualisation', which will show various options for creating new

visualisation, as shown in Figure 8.

Select 'Pie Chart', and it will ask you to 'Select Search Source'. Choose 'From a New Search'. It will display a pie chart and some options to use on the left side. Locate 'Buckets', which will show the option 'Split Slices'. Click on 'Split Slices', then on 'Select Terms in Aggregation', and in 'Field' select 'State' before clicking on the white arrow button within a green square, as shown in Figure 9.

We can see that this has created the pie chart for the city, where the slices are based on the number of students from each city. On the right side, we can see different cities and the colours they represent.

Let's add more to this pie chart. On the left side bottom there is the option to 'Add Sub buckets'. Click on it. Repeat the procedure, i.e., click on 'Split Slice', and then on 'Select Terms in Sub Aggregation'. Then select 'Hobbies' in 'Fields', and click on the white arrow button. You will get what's shown in Figure 10.

Once done, we can save this visualisation for future use, so that the next time we don't have to configure all parameters.

Let us think of other examples, for instance, if we want to know statewise hobbies of students on a gender basis. We can create a data table from the visualisation and can get something like Figure 11.

Once you have data, the possibilities are endless regarding what you can extract from it. The data has all the answers. We only need to ask the right questions. **END** 

References

- [1] <https://lucene.apache.org/core/>
- [2] <https://www.elastic.co/products>
- [3] <https://www.elastic.co/about>
- [4] <https://www.elastic.co/guide>

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